

# Bernstein Clocks for Episodic Continuous-Time MDPs: A Verified Repair Attempt and Partial Progress Report

Prepared for verification of the proposed CT-BEVI manuscript

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## Abstract

This note repairs the parts of the proposed CT-BEVI manuscript that can be made mathematically correct and isolates the parts that still do not prove the open problem of Gao and Zhou. The conclusion is deliberately conservative: the original claimed minimax theorem is not recovered here. What is proved is partial progress. We give corrected one-step Bernstein and rate-sensitivity certificates for the continuous-time Bellman operator, a boundary-compatible clipping operation, and a random-clock Bernstein summation lemma for genuine (on-policy) CTMDP value functions when the jump clock is updated online. These lemmas show that the unbounded random number of jumps is not by itself a fatal obstruction to Bernstein-type estimates. We also prove several no-go statements showing why the submitted proof cannot be fixed by changing constants: its rate confidence operator is false, its transition bonus is not boundary-compatible, and its batch-count summation is false for arbitrary clipped continuation vectors. Thus the paper obtains verified partial progress, not a solution of the Gao–Zhou open problem.

**Status.** This is not a solution to the Gao–Zhou problem. It is a cleaned-up and verified replacement for the incorrect proof attempt. The main missing ingredient for a full solution is an adaptive Bernstein confidence argument for the optimistic value functions generated from the same transition samples, together with an episode-batched random-clock summation. The results below are proved as stated; claims that are only conditional are explicitly labelled as such.

## 1 Model and notation

Let  $\mathcal{S}$  and  $\mathcal{A}$  be finite, with  $|\mathcal{S}| = S$  and  $|\mathcal{A}| = A$ . Write  $z = (x, a)$ . A tabular finite-horizon CTMDP has transition probabilities  $p_z \in \Delta(\mathcal{S})$ , exponential holding-time rates  $\lambda_z \in (0, \Lambda]$ , and known reward rates  $r_z \in [0, 1]$ . An episode has remaining time  $t \in [0, H]$ . If action  $a$  is chosen in state  $x$ , an exponential variable  $E_z \sim \text{Exp}(\lambda_z)$  is drawn. The process spends  $E_z \wedge t$  time units in  $x$ , receives reward  $r_z(E_z \wedge t)$ , and, if  $E_z \leq t$ , jumps to  $Y \sim p_z$  with remaining time  $t - E_z$ .

For  $\lambda > 0$  set

$$\begin{aligned} \tau_\lambda(t) &= \int_0^t e^{-\lambda u} du = \frac{1 - e^{-\lambda t}}{\lambda}, & J_\lambda(t) &= 1 - e^{-\lambda t} = \lambda \tau_\lambda(t), \\ D_\lambda(t) &= \int_0^t u e^{-\lambda u} du = \frac{1 - e^{-\lambda t}(1 + \lambda t)}{\lambda^2}, & q_\lambda(t) &= \int_0^t \lambda e^{-\lambda u}(t - u) du = t - \tau_\lambda(t). \end{aligned}$$

For a vector-valued continuation function  $v = (v_y)_{y \in \mathcal{S}}$  on  $[0, H]$ , define

$$B_v^\lambda(y, t) = \int_0^t \lambda e^{-\lambda u} v_y(t - u) du.$$

The one-step Bellman operator is

$$(\mathcal{T}_z^{\lambda,p}v)(t) = r_z\tau_\lambda(t) + p^\top B_v^\lambda(\cdot, t). \quad (1)$$

Every genuine value function  $V^\pi$  satisfies

$$0 \leq V^\pi(x, t) \leq t, \quad (2)$$

and is 1-Lipschitz in  $t$ . The Lipschitz property follows by coupling two systems with horizons  $t$  and  $t + h$ : the additional reward in the longer system is at most  $h$ .

## 2 Corrected one-step certificates

The original manuscript used bonuses that do not vanish at  $t = 0$  and a rate sensitivity estimate with the wrong clock. This section gives the corrected statements.

### 2.1 A boundary-compatible clipping operator

For a measurable  $h : [0, H] \rightarrow \mathbb{R}$ , define

$$\text{clip}(h)(t) = \min\{t, \max\{0, h(t)\}\}.$$

For  $0 \leq h(t) \leq t$  and  $h(0) = 0$ , define its least 1-Lipschitz majorant by

$$\text{LCh}(t) = \sup_{s \in [0, H]} (h(s) - |t - s|). \quad (3)$$

**Lemma 1** (Boundary-compatible Lipschitz clipping). *If  $0 \leq h(t) \leq t$  for all  $t$  and  $h(0) = 0$ , then  $\text{LCh}$  is 1-Lipschitz,  $0 \leq \text{LCh}(t) \leq t$ ,  $\text{LCh}(0) = 0$ , and  $\text{LCh} \geq h$ . Consequently, if  $h \geq V$  and  $0 \leq V(t) \leq t$ , then  $\text{LC}(\text{clip}h) \geq V$ .*

*Proof.* The supremum of 1-Lipschitz functions is 1-Lipschitz, so  $\text{LCh}$  is 1-Lipschitz. Since the term with  $s = t$  equals  $h(t)$ , we have  $\text{LCh} \geq h$ . The lower bound is then immediate from  $h \geq 0$ . For the upper bound, use  $h(s) \leq s$ :

$$h(s) - |t - s| \leq s - |t - s| \leq t$$

for every  $s, t \in [0, H]$ . At  $t = 0$ ,  $h(s) - s \leq 0$  and the choice  $s = 0$  gives 0, so  $\text{LCh}(0) = 0$ . Finally, if  $h \geq V$  and  $V \leq t$ , then  $\text{clip}h \geq V$  pointwise, and hence  $\text{LC}(\text{clip}h) \geq V$ .  $\square$

### 2.2 Transition range and variance contraction

**Lemma 2** (Range of the continuation vector). *If  $0 \leq v_y(s) \leq s$  for all  $y$  and  $s$ , then*

$$0 \leq B_v^\lambda(y, t) \leq q_\lambda(t) = t - \tau_\lambda(t).$$

*In particular,  $B_v^\lambda(\cdot, 0) = 0$  and the range of  $B_v^\lambda(\cdot, t)$  is at most  $q_\lambda(t)$ .*

*Proof.* The lower bound is clear. For the upper bound,

$$B_v^\lambda(y, t) \leq \int_0^t \lambda e^{-\lambda u} (t - u) du = t - \frac{1 - e^{-\lambda t}}{\lambda} = q_\lambda(t).$$

$\square$

**Lemma 3** (Continuous-time variance contraction). *If  $0 \leq v_y(s) \leq s$ , then for  $g_y = B_v^\lambda(y, t)$ ,*

$$\text{Var}_p(g) \leq J_\lambda(t) \int_0^t \lambda e^{-\lambda u} \text{Var}_p(v.(t-u)) du. \quad (4)$$

*Proof.* Let  $\mu(du) = \lambda e^{-\lambda u} \mathbf{1}\{0 \leq u \leq t\} du$ , whose total mass is  $J_\lambda(t)$ . Denote  $p$ -averages by a bar. Since  $g_y - \bar{g} = \int_0^t (v_y(t-u) - \overline{v(t-u)}) \mu(du)$ , Cauchy's inequality for the finite measure  $\mu$  gives

$$(g_y - \bar{g})^2 \leq J_\lambda(t) \int_0^t (v_y(t-u) - \overline{v(t-u)})^2 \mu(du).$$

Averaging over  $y \sim p$  proves the claim.  $\square$

### 2.3 Rate sensitivity

**Lemma 4** (Correct rate sensitivity). *Assume  $v_y(0) = 0$  and each  $v_y$  is 1-Lipschitz. Then for every  $p \in \Delta(\mathcal{S})$ ,  $r \in [0, 1]$ , and  $\lambda, \lambda' > 0$ ,*

$$|(\mathcal{T}_z^{\lambda, p} v)(t) - (\mathcal{T}_z^{\lambda', p} v)(t)| \leq 2|\lambda - \lambda'| D_{\lambda \wedge \lambda'}(t). \quad (5)$$

*Equivalently, the correct clock is the largest  $D_\mu(t)$  on the interval between  $\lambda$  and  $\lambda'$ , not necessarily  $D_\lambda(t)$  or  $D_{\lambda'}(t)$ .*

*Proof.* Let  $h(u) = p^\top v(t-u)$  for  $u \in [0, t]$ . Then  $h(t) = 0$  and  $h$  is 1-Lipschitz, hence absolutely continuous. Integration by parts gives

$$\int_0^t \lambda e^{-\lambda u} h(u) du = h(0) + \int_0^t e^{-\lambda u} h'(u) du,$$

where  $|h'(u)| \leq 1$  a.e. Therefore

$$\left| \frac{\partial}{\partial \lambda} \int_0^t \lambda e^{-\lambda u} h(u) du \right| = \left| \int_0^t -u e^{-\lambda u} h'(u) du \right| \leq D_\lambda(t).$$

Also  $|\partial \tau_\lambda(t)/\partial \lambda| = D_\lambda(t)$ . The derivative of (1) with respect to  $\lambda$  is therefore bounded in absolute value by  $2D_\lambda(t)$ . Since  $D_\mu(t)$  is decreasing in  $\mu$ , integration from  $\lambda$  to  $\lambda'$  gives (5).  $\square$

### 2.4 Fixed-query Bernstein transition certificates

The next lemma is intentionally stated only for fixed or conditionally fixed queries. It is not valid for an arbitrary vector chosen from the same samples unless one adds a sample-splitting, stability, or uniform-complexity argument.

**Lemma 5** (Fixed-query Bernstein certificate). *Let  $Y_1, \dots, Y_n$  be i.i.d. with law  $p$  on  $\mathcal{S}$ , let  $\hat{p}_n$  be the empirical law, and let  $g \in [0, R]^S$  be deterministic conditional on a sigma-field independent of  $Y_1, \dots, Y_n$ . Then, for every  $\ell > 0$ , with conditional probability at least  $1 - 2e^{-\ell}$ ,*

$$|(\hat{p}_n - p)^\top g| \leq \sqrt{\frac{2 \text{Var}_p(g) \ell}{n}} + \frac{2R\ell}{3n}. \quad (6)$$

*Moreover, standard empirical Bernstein inequalities give the same form with  $\text{Var}_p(g)$  replaced by the empirical variance, at the cost of changing the absolute constants.*

*Proof.* Condition on the sigma-field with respect to which  $g$  is fixed. The variables  $g(Y_i) - p^\top g$  are independent, mean zero, bounded by  $R$  in absolute value, and have variance  $\text{Var}_p(g)$ . Bernstein's inequality gives (6). The empirical-variance version is the standard empirical Bernstein inequality of Maurer and Pontil.  $\square$

**Corollary 1** (A boundary-compatible local transition bonus). *For a query of the form  $g = B_v^\lambda(\cdot, t)$  with  $0 \leq v_y(s) \leq s$ , the range term in a Bernstein bonus can be taken proportional to  $q_\lambda(t)\ell/n$ , not  $H\ell/n$ . In particular, the bonus vanishes at  $t = 0$ .*

*Proof.* Apply Lemma 2 with  $R = q_\lambda(t)$  in Lemma 5.  $\square$

## 2.5 A mathematically consistent one-step optimistic backup

A corrected one-step optimistic backup should use a rate bonus based on Lemma 4 and a transition bonus based on Corollary 1. If  $\mathcal{I}_k(z)$  is a valid confidence interval for  $\lambda_z$  and  $\underline{\lambda}_k(z) = \inf \mathcal{I}_k(z)$ , then, with the convention  $D_0(t) = t^2/2$  if the lower endpoint is zero, for every  $\lambda \in \mathcal{I}_k(z)$ ,

$$|(\mathcal{T}_z^{\lambda, p_z} v)(t) - (\mathcal{T}_z^{\lambda_z, p_z} v)(t)| \leq 2|\lambda - \lambda_z| D_{\underline{\lambda}_k(z)}(t). \quad (7)$$

Thus a safe rate bonus is  $2\alpha_k(z)D_{\underline{\lambda}_k(z)}(t)$  whenever  $|\lambda - \lambda_z| \leq \alpha_k(z)$  on the confidence event.

The raw optimistic backup may then be clipped by  $\text{LC}(\text{clip}(\cdot))$ . This clipping is not an upper projection of the raw expression; it is the correct projection for optimism because genuine values are bounded above by  $t$ . If the raw expression is above the true Bellman value, clipping at  $t$  and then applying LC preserves that inequality by Lemma 1.

## 3 A verified Bernstein clock lemma for genuine values

Gao and Zhou emphasize that the random and unbounded number of decision epochs is a central obstruction to importing Bernstein bonuses from discrete time. The following proposition shows that this obstruction can be overcome for genuine on-policy value functions when the jump counter is updated online, i.e., after each completed jump rather than only between episodes. This does not yet prove the desired CT-BEVI theorem, because CT-BEVI uses optimistic values generated from data and, in the original episodic protocol, uses episode-start statistics.

Fix an arbitrary learning rule that, at the beginning of episode  $k$ , chooses a nonanticipating policy  $\pi_k$ . Let  $V_k = V_M^{\pi_k}$  be the true continuation value of that policy in the true CTMDP. At a decision epoch in episode  $k$ , let  $Z_{k,j}$  be the chosen state-action pair and  $T_{k,j}$  the remaining time. Let  $N_{k,j}^-(z)$  denote the number of completed jumps from pair  $z$  before this decision epoch, counting jumps earlier in the same episode. Define

$$g_{k,j}(y) = B_{V_k}^{\lambda_{Z_{k,j}}}(y, T_{k,j}).$$

**Proposition 1** (Online jump-clock Bernstein summability). *For every CTMDP with rates bounded by  $\Lambda$  and rewards in  $[0, 1]$ ,*

$$\mathbb{E} \sum_{k=1}^K \sum_{j: T_{k,j} > 0} \sqrt{\frac{\text{Var}_{p_{Z_{k,j}}}(g_{k,j})}{N_{k,j}^-(Z_{k,j}) \vee 1}} \leq H \sqrt{SAK(1 + \log(1 + K\Lambda H))}. \quad (8)$$

*Up to constants and logarithms, this is the desired  $H\sqrt{SAK}$  Bernstein scaling for the transition-variance part.*

*Proof.* Fix a state-action pair  $z$  and enumerate all decision epochs at which  $Z_{k,j} = z$  by  $i = 1, 2, \dots$ . Let  $M_i$  be the number of completed jumps from  $z$  before the  $i$ th such decision, including jumps earlier in the same episode. Let  $t_i$  be the remaining time and let

$$J_i = J_{\lambda_z}(t_i) = 1 - e^{-\lambda_z t_i}$$

be the conditional probability that the decision produces a completed jump. Let  $\xi_i$  be the completed-jump indicator, so  $\mathbb{E}[\xi_i \mid \mathcal{F}_i] = J_i$  and  $M_{i+1} = M_i + \xi_i$  for the subsequence of visits to  $z$ . Finally set

$$W_i = \int_0^{t_i} \lambda_z e^{-\lambda_z u} \text{Var}_{p_z}(V_k(\cdot, t_i - u)) du.$$

By Lemma 3,  $\text{Var}_{p_z}(g_i) \leq J_i W_i$ . Hence, for this fixed  $z$ ,

$$\sum_i \sqrt{\frac{\text{Var}_{p_z}(g_i)}{M_i \vee 1}} \leq \left( \sum_i W_i \right)^{1/2} \left( \sum_i \frac{J_i}{M_i \vee 1} \right)^{1/2}. \quad (9)$$

The second factor is controlled by the jump clock. Since  $\mathbb{E}[\xi_i / (M_i \vee 1) \mid \mathcal{F}_i] = J_i / (M_i \vee 1)$ ,

$$\mathbb{E} \sum_i \frac{J_i}{M_i \vee 1} = \mathbb{E} \sum_i \frac{\xi_i}{M_i \vee 1} \leq 1 + \mathbb{E} \log(1 + M_\infty) \leq 1 + \log(1 + K\Lambda H),$$

where  $M_\infty$  is the total number of completed jumps from  $z$  over all  $K$  episodes. The last inequality uses Jensen and  $\mathbb{E} M_\infty \leq K\Lambda H$ , obtained by domination by a Poisson process of rate  $\Lambda$  over total time  $KH$ .

It remains to sum the  $W_i$  terms. In episode  $k$ , let  $R_k$  be the total future reward obtained under policy  $\pi_k$ ; then  $0 \leq R_k \leq H$ . Consider the Doob martingale of  $R_k$  generated by revealing, at each decision, the holding time and then the next-state mark when a jump occurs. The conditional variance contribution of the next-state mark at a decision epoch with pair  $z$  and remaining time  $t$  is

$$\mathbf{1}\{E_z \leq t\} \text{Var}_{p_z}(V_k(\cdot, t - E_z)).$$

Taking conditional expectation over the holding time gives precisely the corresponding  $W_i$ . Since the sum of the conditional variances of selected martingale increments is bounded by the total conditional variance of  $R_k$ , we have

$$\mathbb{E} \left[ \sum_j W_{k,j} \mid \mathcal{F}_{k,0} \right] \leq \text{Var}(R_k \mid \mathcal{F}_{k,0}) \leq H^2/4.$$

Therefore  $\sum_z \mathbb{E} A_z \leq KH^2/4$ , where  $A_z = \sum_{i: Z_i=z} W_i$ . Taking expectations in (9), applying Cauchy's inequality, and summing over  $z$  gives

$$\begin{aligned} \mathbb{E} \sum_{k,j} \sqrt{\frac{\text{Var}_{p_{Z_{k,j}}}(g_{k,j})}{N_{k,j}^-(Z_{k,j}) \vee 1}} &\leq \sum_z \sqrt{\mathbb{E} A_z \mathbb{E} B_z} \\ &\leq \sqrt{\left( \sum_z \mathbb{E} A_z \right) \left( \sum_z \mathbb{E} B_z \right)} \\ &\leq H \sqrt{SAK(1 + \log(1 + K\Lambda H))}, \end{aligned}$$

where  $B_z = \sum_{i: Z_i=z} J_i / (M_i \vee 1)$ . This proves the claim, with a harmless absolute-constant slack in the displayed bound.  $\square$

**Remark 1** (Why this is only partial progress). *Proposition 1 applies to genuine continuation values  $V_k = V_M^{\pi_k}$  and to an online-updated jump counter. The proposed CT-BEVI proof needs a corresponding statement for optimistic values computed from the same empirical transition law and for statistics frozen at the start of an episode. Those two changes are substantial. They are exactly where the submitted proof had gaps.*

## 4 Exposure-clock algebra

The rate-estimation part should be paid for in exposure time. The following simple deterministic inequalities are often the right replacement for a pigeonhole count.

**Lemma 6** (Exposure-clock inequalities). *Let  $d_i \geq 0$ , let  $S_i = \sum_{\ell < i} d_\ell$ , and let  $S_{m+1} = \sum_{i=1}^m d_i$ . Terms with  $d_i = 0$  are interpreted as zero. Then, for every  $\eta > 0$ ,*

$$\sum_{i=1}^m \frac{d_i}{\sqrt{S_i + d_i}} \leq 2\sqrt{S_{m+1}}, \quad (10)$$

$$\sum_{i=1}^m \frac{d_i}{S_i + d_i + \eta} \leq \log \left( 1 + \frac{S_{m+1}}{\eta} \right). \quad (11)$$

*Proof.* For (10), use  $d_i/(\sqrt{S_i + d_i}) \leq 2(\sqrt{S_i + d_i} - \sqrt{S_i})$  and telescope. For (11), use

$$\frac{d_i}{S_i + d_i + \eta} \leq \int_{S_i}^{S_i + d_i} \frac{dx}{x + \eta}.$$

The inequality holds because  $x + \eta \leq S_i + d_i + \eta$  on the integration interval. Summing over  $i$  telescopes the logarithms.  $\square$

**Remark 2.** *For a rate- $\lambda$  holding time censored at remaining time  $t$ , the expected exposure is  $\tau_\lambda(t)$ , and Lemma 4 involves  $D_\lambda(t)$ . The useful algebraic relation is*

$$\frac{\lambda D_\lambda(t)^2}{\tau_\lambda(t)} \leq t \tau_\lambda(t). \quad (12)$$

*Indeed,  $D_\lambda(t) \leq \tau_\lambda(t) \min\{t, 1/\lambda\}$ , which implies (12). This is a verified ingredient for a rate-Bernstein analysis, but by itself it does not prove a regret theorem: a valid proof must still handle confidence intervals whose lower endpoint is far below the true rate and must relate expected exposure to the denominator used by the actual algorithm.*

## 5 No-go statements for the submitted proof

The previous sections state what can be repaired. This section records the main points that cannot be repaired by changing constants.

**Proposition 2** (The original boundary condition is impossible). *The transition bonus used in the submitted manuscript has the form*

$$6\sqrt{\frac{\widehat{\text{Var}}_{k,z}(g)L}{N_k(z) \vee 1}} + \frac{18HL}{N_k(z) \vee 1}.$$

*At  $t = 0$ ,  $g = B_v^\lambda(\cdot, 0) = 0$ , so the square-root term vanishes but the additive term is positive. Therefore a projection that is required both to vanish at  $t = 0$  and to lie above the raw backup cannot exist.*

*Proof.* At  $t = 0$  the raw backup is at least  $18HL/(N_k(z) \vee 1) > 0$ . Any function that lies above it has positive value at 0, while the proposed clipped value is required to equal 0 at 0. These requirements are incompatible.  $\square$

**Proposition 3** (The rate-confidence operator in the manuscript is false). *No universal constant  $C$  can make the bound*

$$|(\mathcal{T}_z^{\lambda', p}v)(t) - (\mathcal{T}_z^{\lambda, p}v)(t)| \leq C|\lambda' - \lambda|D_\lambda(t)$$

*hold for all  $\lambda' \leq \lambda$ , even with  $v = 0$  and  $r = 1$ .*

*Proof.* Take  $H = t = 1$ ,  $v = 0$ ,  $r = 1$ ,  $\lambda = \Lambda$ , and  $\lambda' = \varepsilon$ . Then

$$(\mathcal{T}_z^{\lambda, p}0)(1) = \tau_\lambda(1).$$

As  $\varepsilon \downarrow 0$  and  $\Lambda \rightarrow \infty$ ,

$$|\tau_\varepsilon(1) - \tau_\Lambda(1)| \rightarrow 1, \quad |\Lambda - \varepsilon|D_\Lambda(1) \leq \frac{\Lambda}{\Lambda^2} = \frac{1}{\Lambda} \rightarrow 0.$$

Thus the proposed inequality fails. Lemma 4 shows the correct replacement: one must use  $D_{\lambda \wedge \lambda'}(t)$ .  $\square$

**Proposition 4** (A batch-count Bernstein summation is false for arbitrary clipped queries). *There is no bound of the form*

$$CH\sqrt{SAK} \text{polylog}(S, A, K, \Lambda H)$$

*for*

$$\mathbb{E} \sum_{k,j} \sqrt{\frac{\text{Var}_{pZ_{k,j}}(g_{k,j})}{N_k(Z_{k,j}) \vee 1}}$$

*when  $N_k$  is frozen at the start of each episode and the vectors  $g_{k,j} \in [0, H]^S$  are allowed to be arbitrary predictable clipped vectors.*

*Proof.* It suffices to take  $K = 1$ , two states, one action per state, rate  $\Lambda$  at each state, and transition law  $p = (1/2, 1/2)$  from each state. At the beginning of the only episode,  $N_1(z) = 0$  for both state-action pairs. Let  $g_j = (0, H)$  at every decision epoch, so  $\text{Var}_p(g_j) = H^2/4$  for whichever pair is visited. Because the rate is always  $\Lambda$ , the number of completed jumps by time  $H$  is Poisson with mean  $\Lambda H$ , and the number of decision epochs is one plus this count. Therefore

$$\mathbb{E} \sum_j \sqrt{\text{Var}_p(g_j)/(N_1(Z_j) \vee 1)} = \frac{H}{2}(1 + \Lambda H) \geq c\Lambda H^2$$

for a numerical constant  $c > 0$ . This exceeds any bound with only logarithmic dependence on  $\Lambda H$  as  $\Lambda \rightarrow \infty$ .  $\square$

**Remark 3** (The adaptive empirical-Bernstein gap). *Lemma 5 is valid for fixed or independently chosen queries. In CT-BEVI, however, the vector  $B_{V_k}^\lambda(\cdot, t)$  is computed from the same empirical transition law  $\hat{p}_k$  that appears in the deviation  $(\hat{p}_k - p)^\top B_{V_k}^\lambda$ . A fixed-query Bernstein inequality does not justify this step. One needs sample splitting, a stability argument, or a uniform confidence set for the class of possible optimistic value vectors. The uploaded manuscript supplies none of these.*

## 6 What would still be needed for a full solution

The following conditional statement summarizes the remaining route. It is not claimed as a theorem about CT-BEVI; it is a checklist of the missing pieces.

**Proposition 5** (Conditional route to the desired rate). *Suppose an implementable optimistic CT-MDP algorithm admits, on an event of probability at least  $1 - \delta$ , a regret decomposition of the form*

$$\text{Reg}(K) \leq C \sum_{k,j} \sqrt{\frac{\text{Var}_{p_{Z_{k,j}}}(G_{k,j})L}{\mathcal{N}_{k,j}(Z_{k,j}) \vee 1}} + C \sum_{k,j} R_{k,j}^\lambda + CHSAL^2,$$

where  $L$  is logarithmic,  $G_{k,j}$  are continuation vectors for which the on-policy variance-clock argument of Proposition 1 is valid, the counters  $\mathcal{N}_{k,j}$  are online jump counters or an equivalent episode-batched substitute, and the rate terms  $R_{k,j}^\lambda$  are payable by the exposure-clock inequalities of Lemma 6 together with (12). Then the leading transition and rate terms are  $\tilde{O}(H\sqrt{SAK})$ .

*Proof.* This is immediate from Proposition 1, the exposure-clock inequalities, and Cauchy’s inequality over the  $SA$  state-action clocks. The statement is conditional because the hypotheses encode exactly the adaptive optimism and episode-batching issues not solved in this note.  $\square$

## 7 Conclusion

The submitted CT-BEVI manuscript cannot be repaired into a proof of the claimed sharp minimax regret theorem without new ideas. The corrected statements in this note constitute partial progress: the local Bellman analysis is fixed, the right rate clock is identified, the transition bonus can be made compatible with  $t = 0$ , and a genuine-value online-clock Bernstein summation is proved. These results support the plausibility of a Bernstein approach in continuous time, but they do not solve the Gao–Zhou open problem. The two unresolved issues are:

- (i) a valid adaptive empirical-Bernstein confidence argument for optimistic value functions computed from the same transition data; and
- (ii) a replacement for the online jump-clock proof that works with statistics frozen at episode boundaries, or a reformulation of the learning protocol that legitimately updates the policy and counters inside an episode.

Until those issues are solved, the sharp  $\tilde{O}(H\sqrt{SAK})$  CTMDP regret bound remains unproved by this line of argument.

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